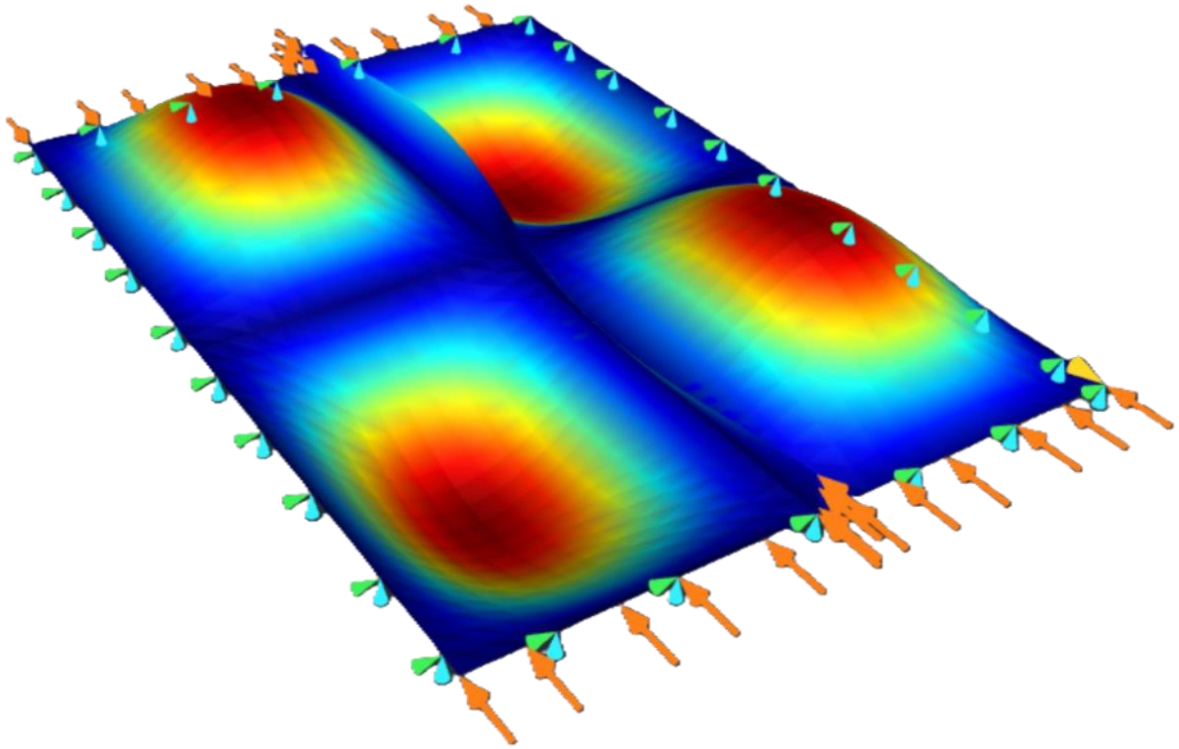
ADVANCED AIRCRAFT DESIGN

LINEAR FINITE ELEMENTS BUCKLING EXERCISES





INTRODUCTION

- The purpose of this exercise is to get exposure to a representative sizing problem for **light structures**

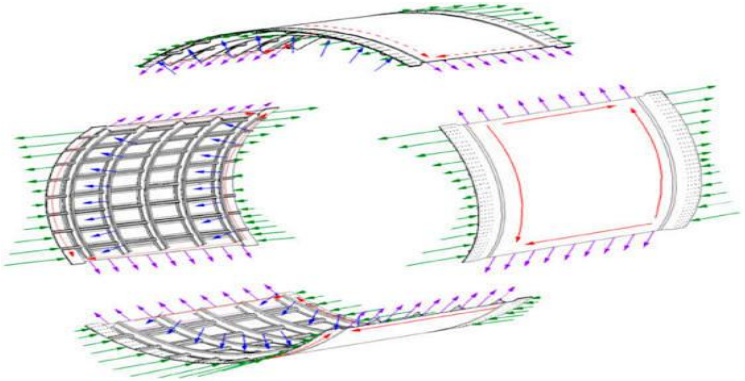
- A **stiffened panel,** with two levels of curvature (one representative of a fuselage and the other effectively a flat plate) will be "sized" to prevent buckling for a given set of loads

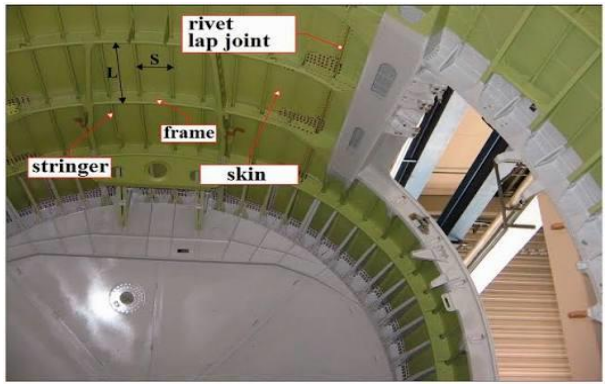
- The panel will be sized both using isotropic (e.g. metal) and laminated (composites) materials.

- The trends relating panel total weight and buckling eigenvalues will be plotted and compared with the theoretical predictions

- Additionally, a critique of the structural model (loads and boundary conditions) will be required

- The first task is to install the Julia language (in which the Finite Elements Solver has been written) and the solver itself, follow the instructions in the next pages

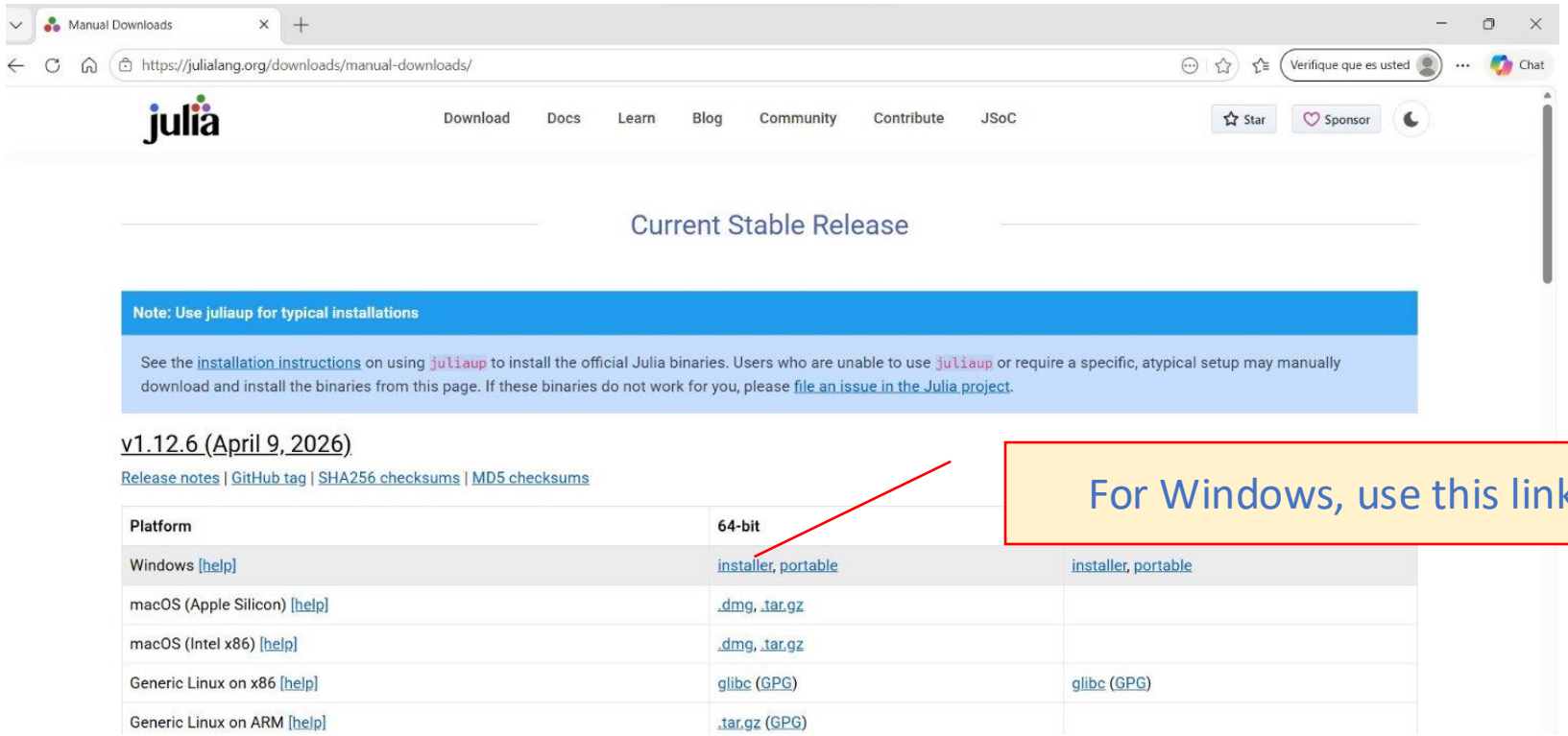




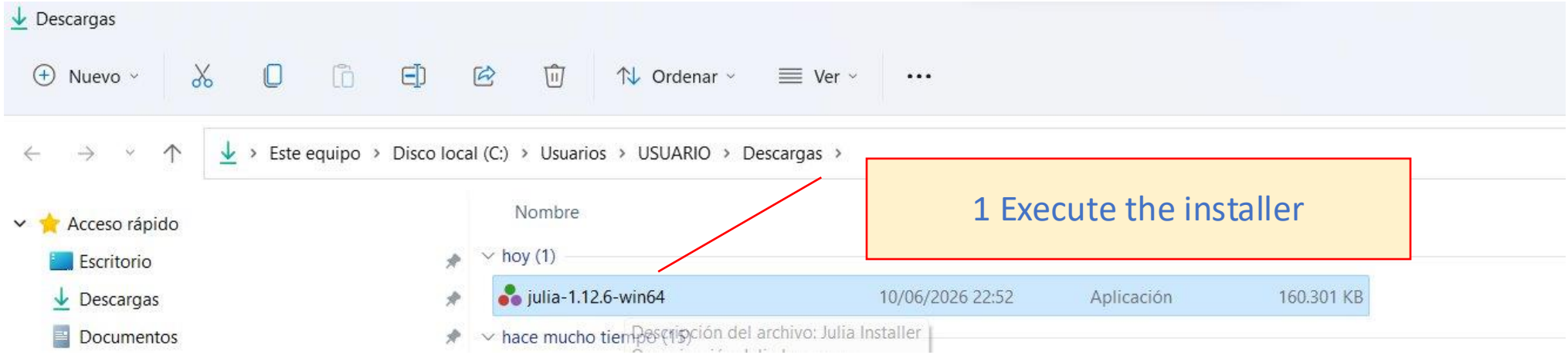


INSTALL THE JULIA LANGUAGE

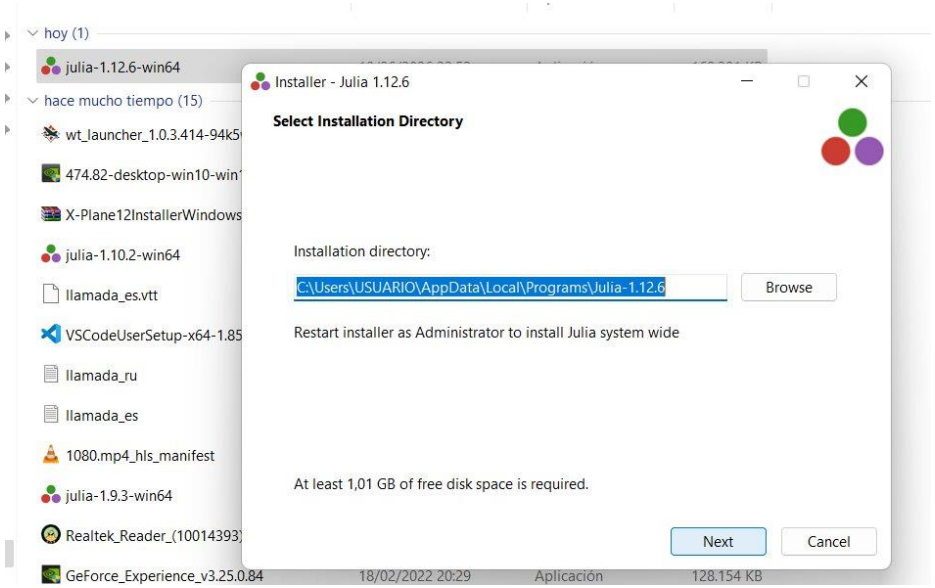
- Go to Manual Downloads (https://julialang.org/downloads/manual-downloads/)
- Select the installer for your system (Windows, Mac or Linux) and download it
- If you are curious about Julia check this Mathematical_tourism_with_Julia/COURSE_PRESENTATIONS/Julia Course_v18.pdf at main · flt-acdesign/Mathematical_tourism_with_Julia



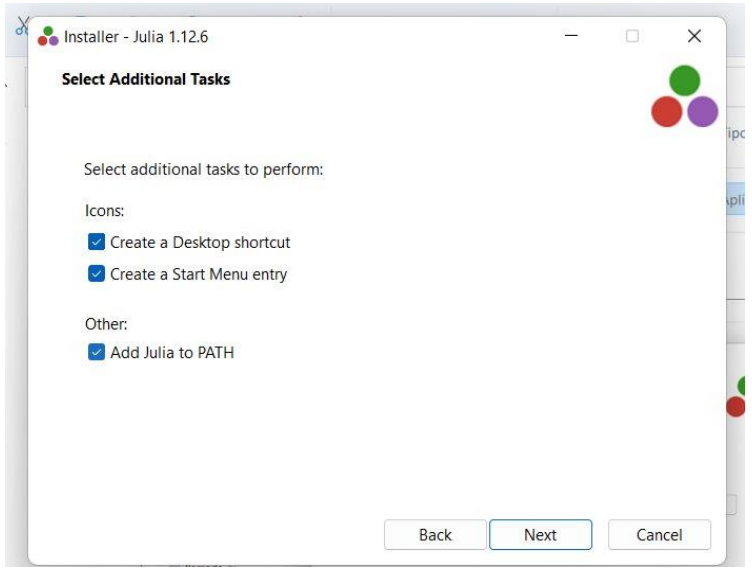
JULIA INSTALLATION PROCESS



1 Execute the installer



2 Choose an installation folder



3 Click on "Add Julia to PATH" (very important)

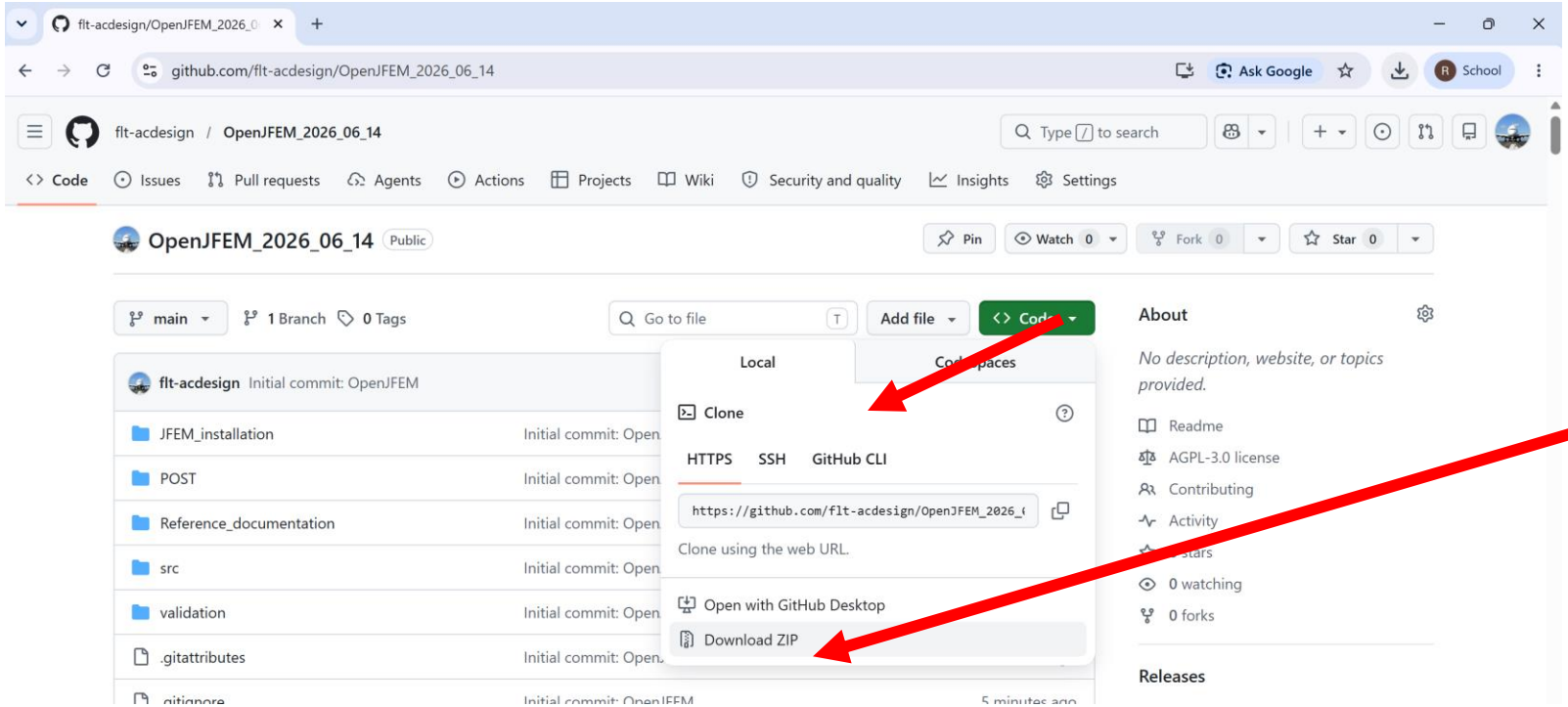


4 Julia is installed, no need to do anything here, close the window

ACCESS THE OpenJFEM REPOSITORY FROM GitHub

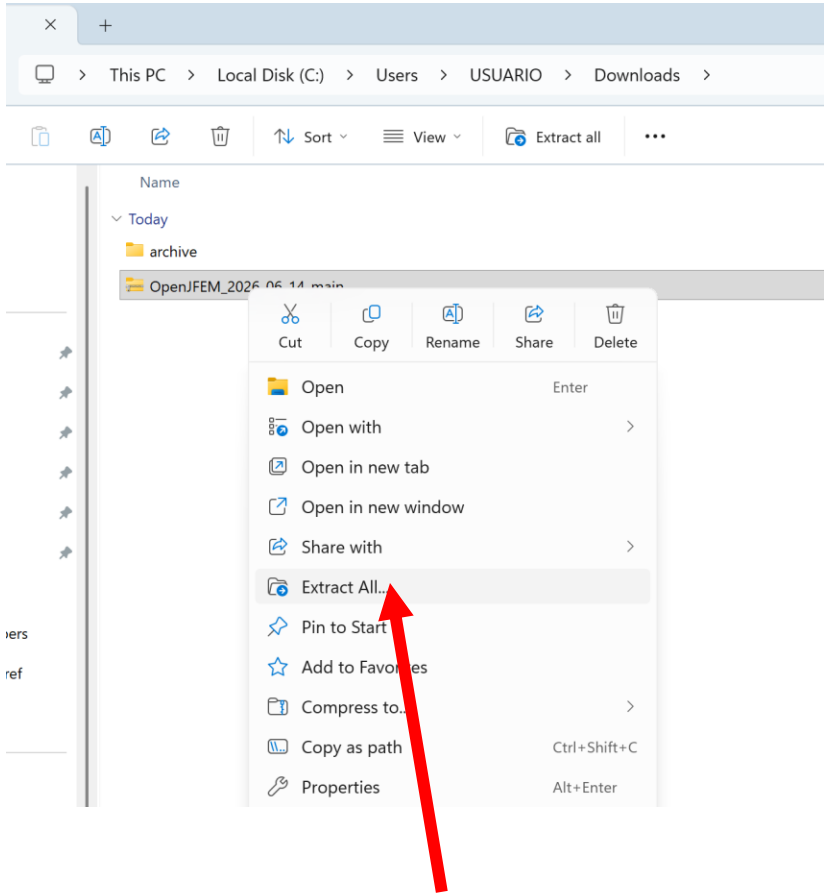
1 Access the GitHub OpenJFEM repository in this link: [flt-acdesign/OpenJFEM_2026_06_14]

Or use this URL https://github.com/flt-acdesign/OpenJFEM_2026_06_14

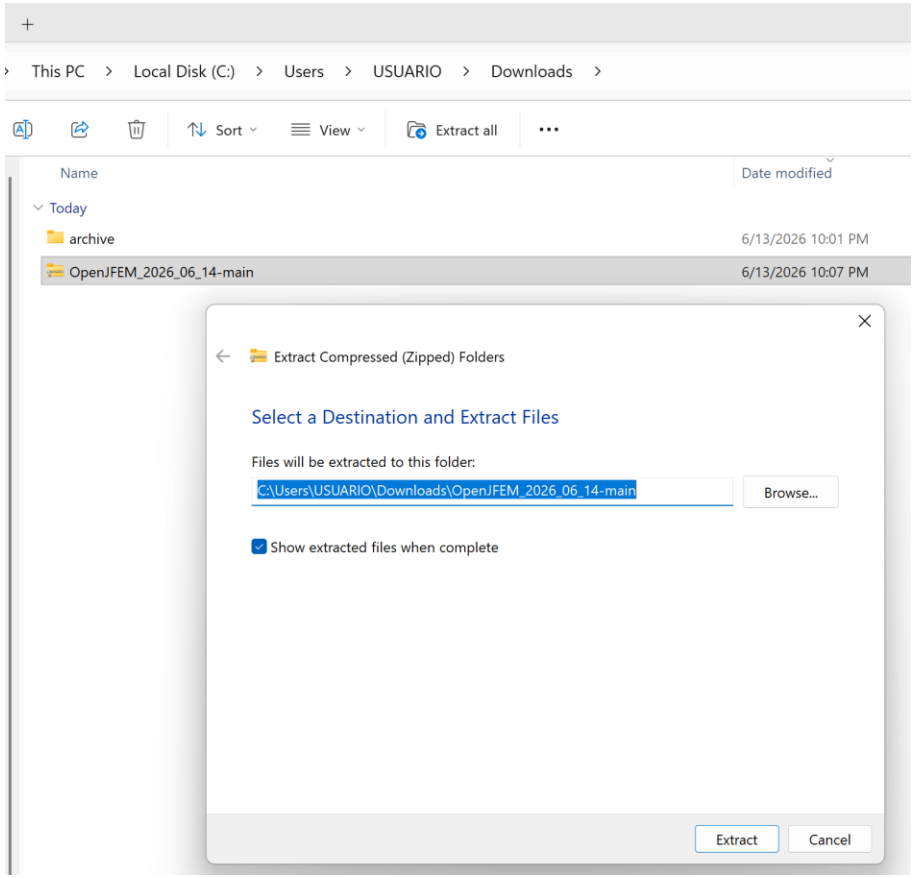


2 Download the repository as a Zip file

SAVE THE REPOSITORY AND UNCOMPRESS IT

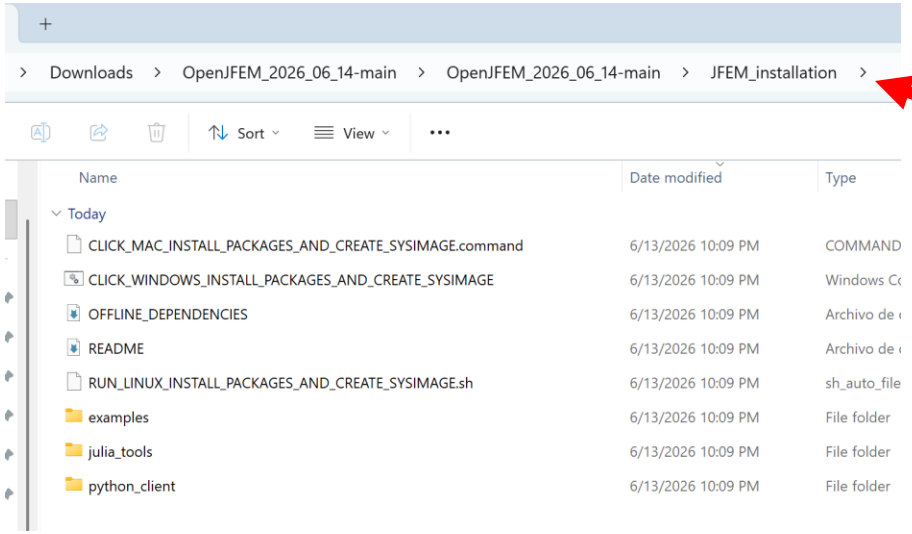


1 Expand the downloaded repository Zip file

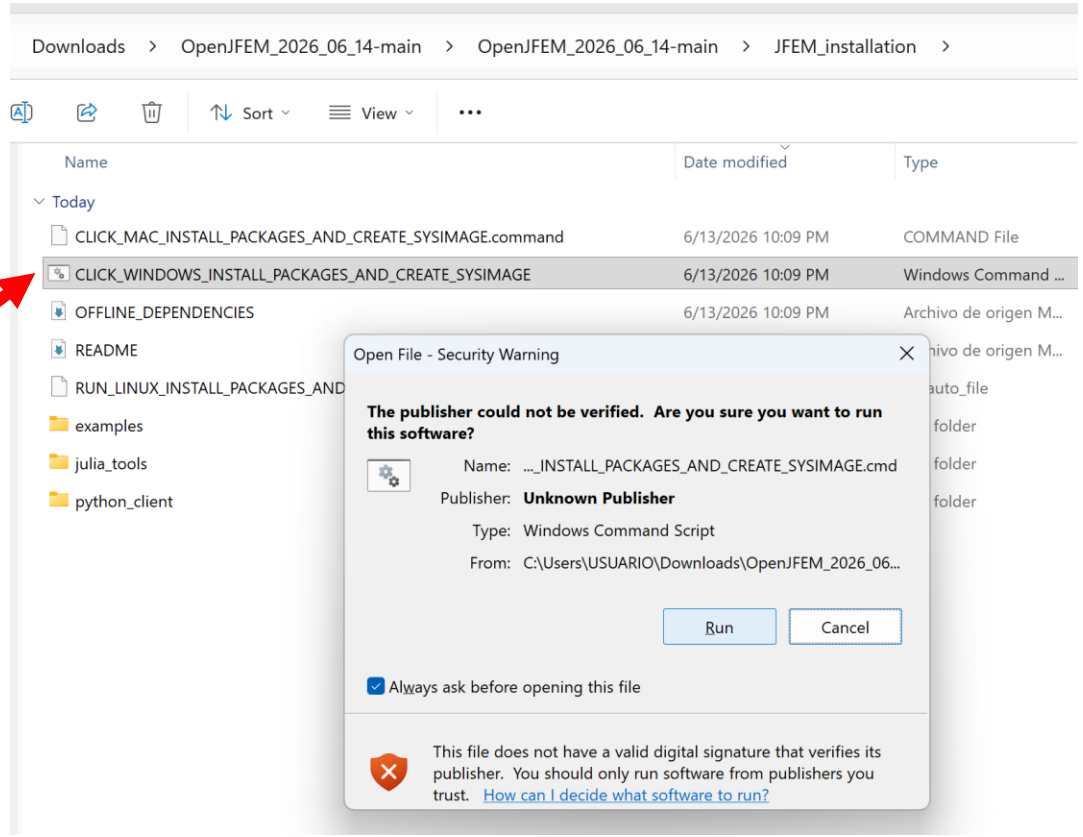


2 Select a folder for the uncompressed repository (it does not matter; you can keep it in "Downloads"

INSTALL THE REQUIRED JULIA PACKAGES AND CREATE A "SYSIMAGE"



1 Navigate to this folder in the expanded repository folder



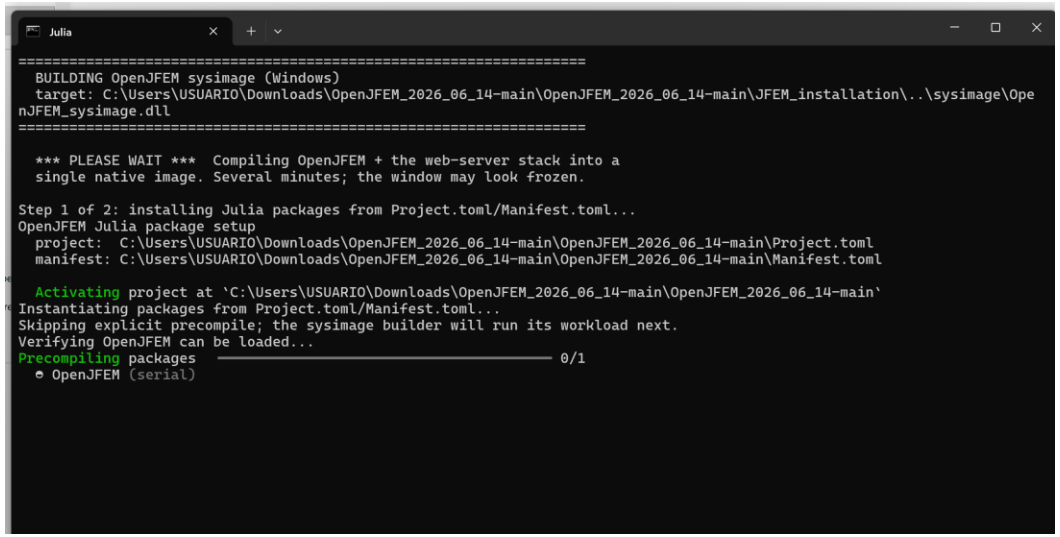
2 If you are in Windows, double click on the file highlighted and press "RUN"

This will install the required Julia packages (libraries) and create a DLL file for your computer)

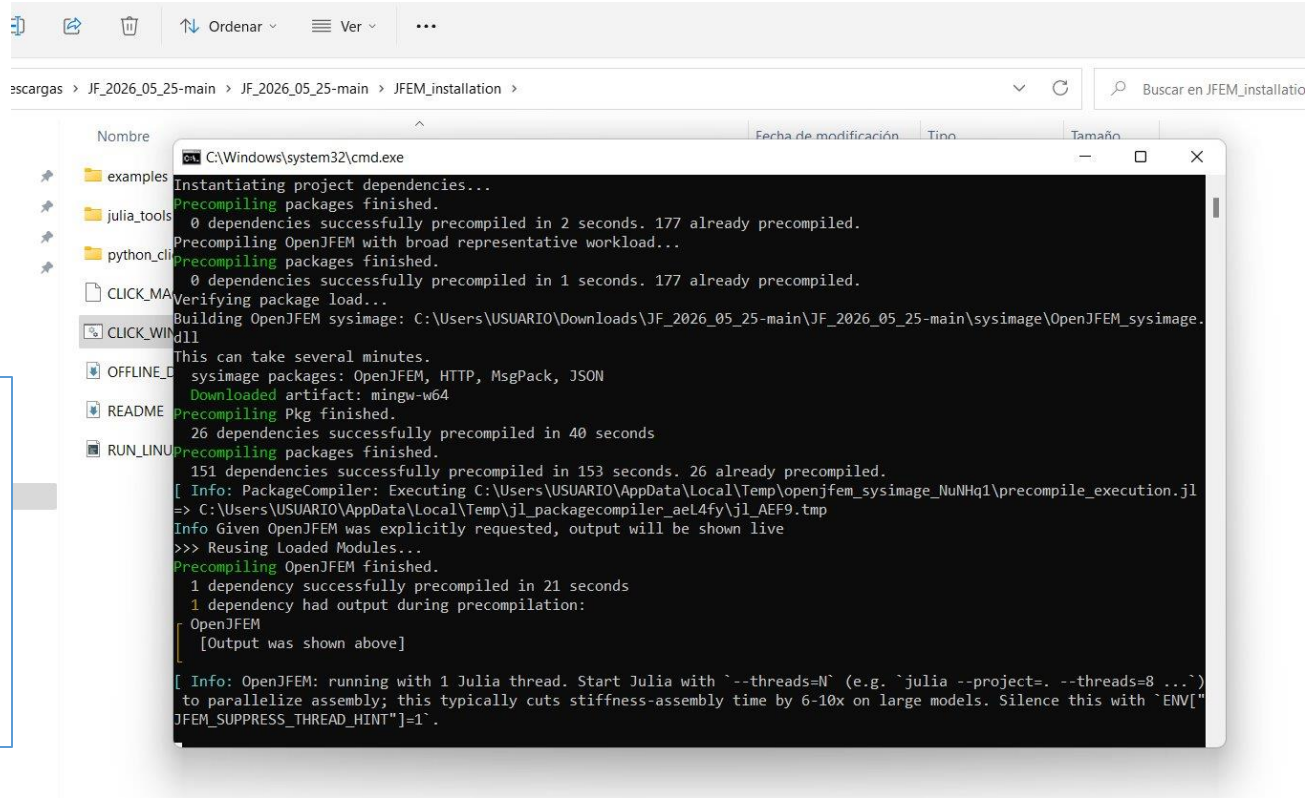
THIS CAN TAKE UP TO 45 MINUTES!

If you are using Linux or Mac, execute the last file in this list

INSTALLING THE PACKAGES AND CREATING THE SYSIMAGE



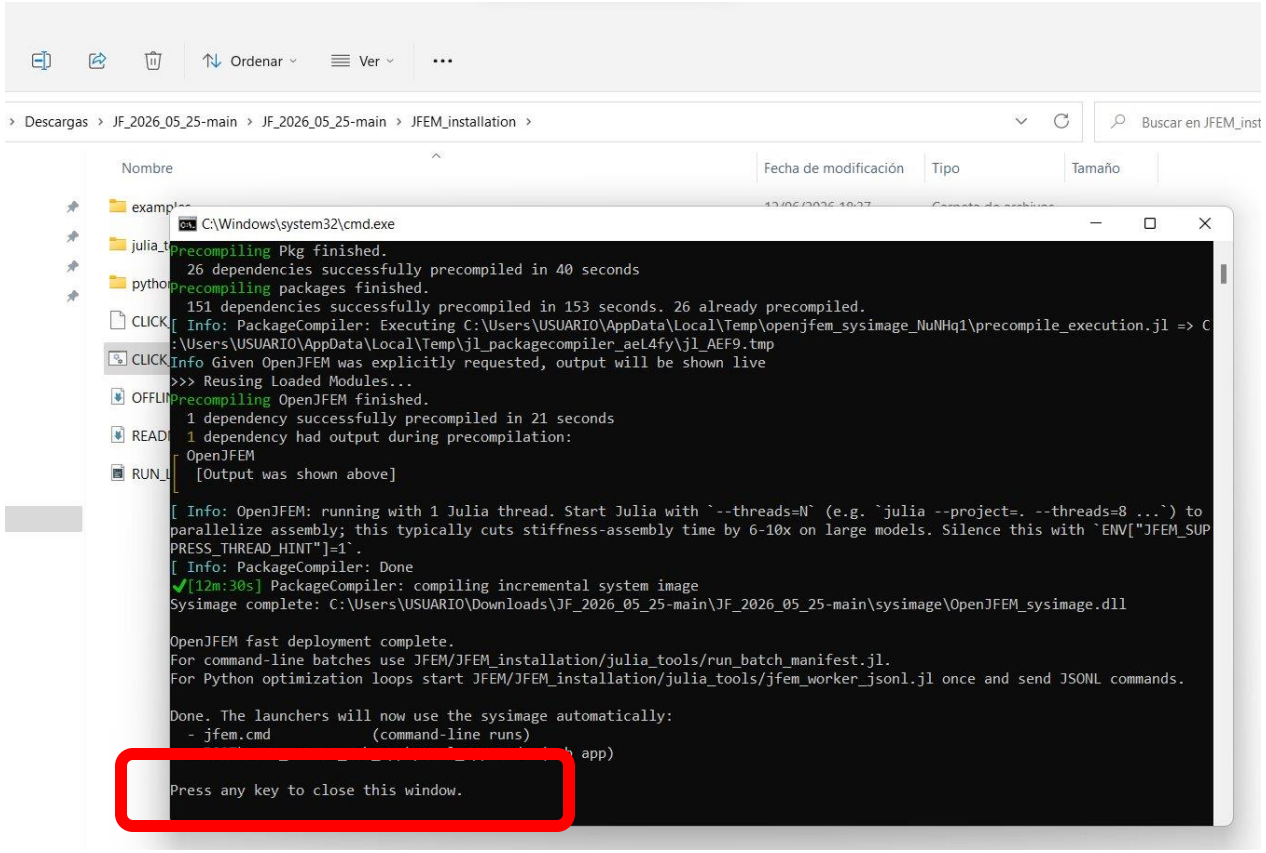
Accept the execution of the command file, this will launch various processes



The installation can take a long time (up to 45 minutes on an "old" computer). Be patient and do something else meanwhile.

You need to be connected to the internet as Julia will download the packages (libraries).

INSTALLATION COMPLETE



Once you see the "Press any key to close this window" message, you have completed the installation. Click on the black screen and press any key, it should close

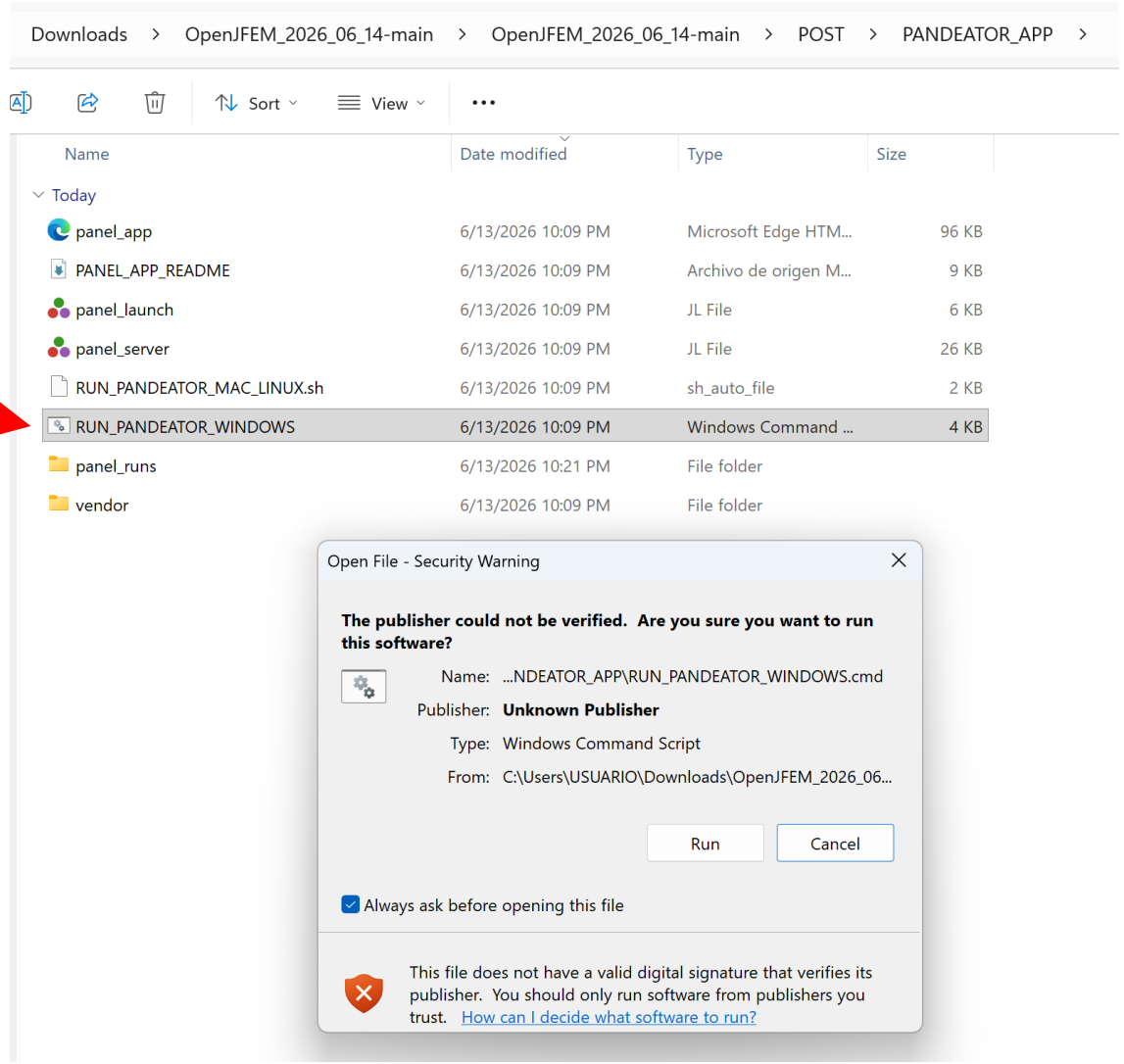
RUNNING PANDEATOR (THE JULIA CODE TO ANALYZE STIFFENED PANELS)

Navigate to this folder and execute the corresponding file "RUN_PANDEATOR_***", where "***" is the name of your operating system

For windows, double click on

and click "Run"

For Mac or Linux execute the corresponding file

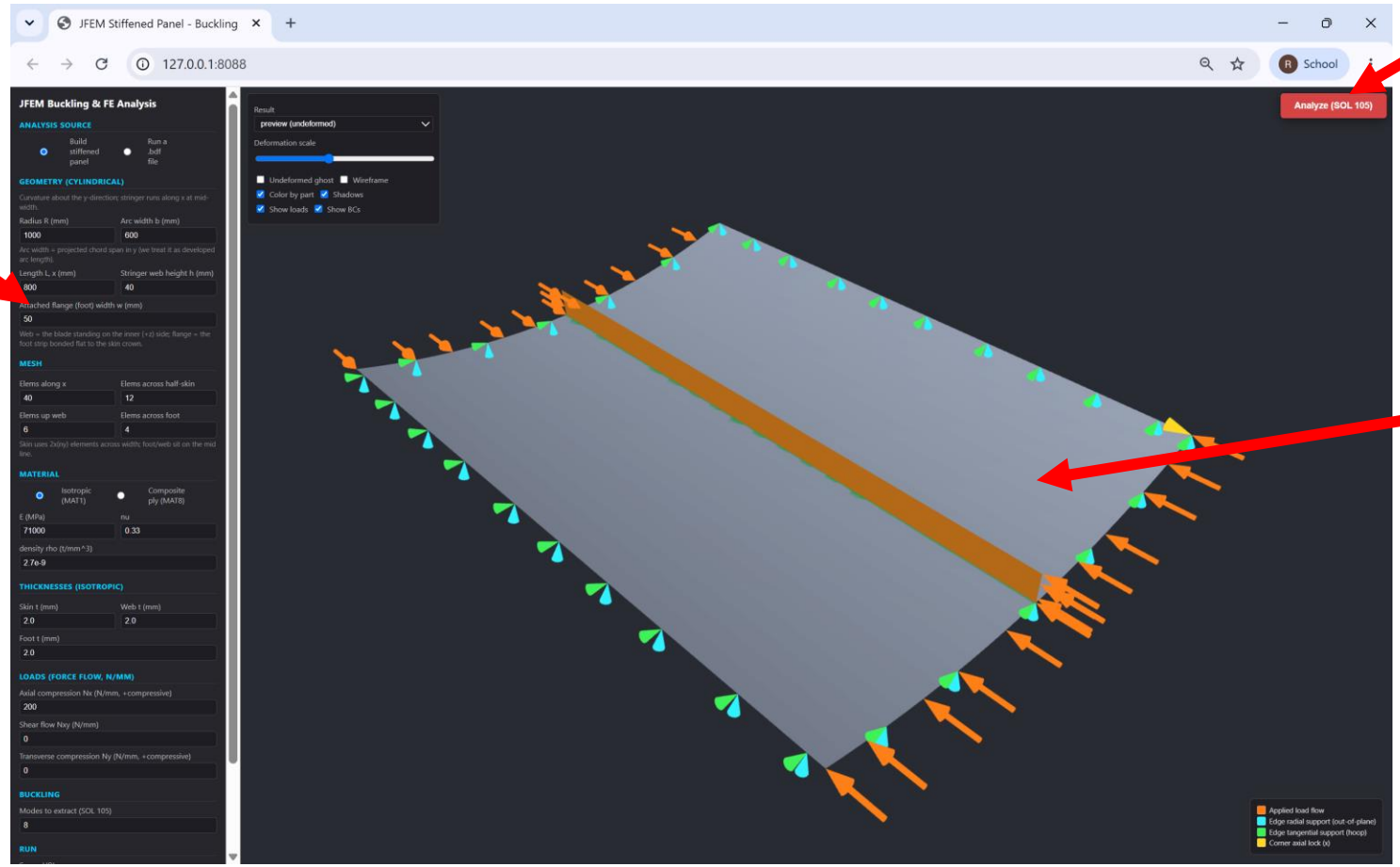


GRAPHICAL USER'S INTERFACE (GUI)

The code will launch your browser and you will see the GUI below.

Geometric parameters, material and loads

Run the analysis

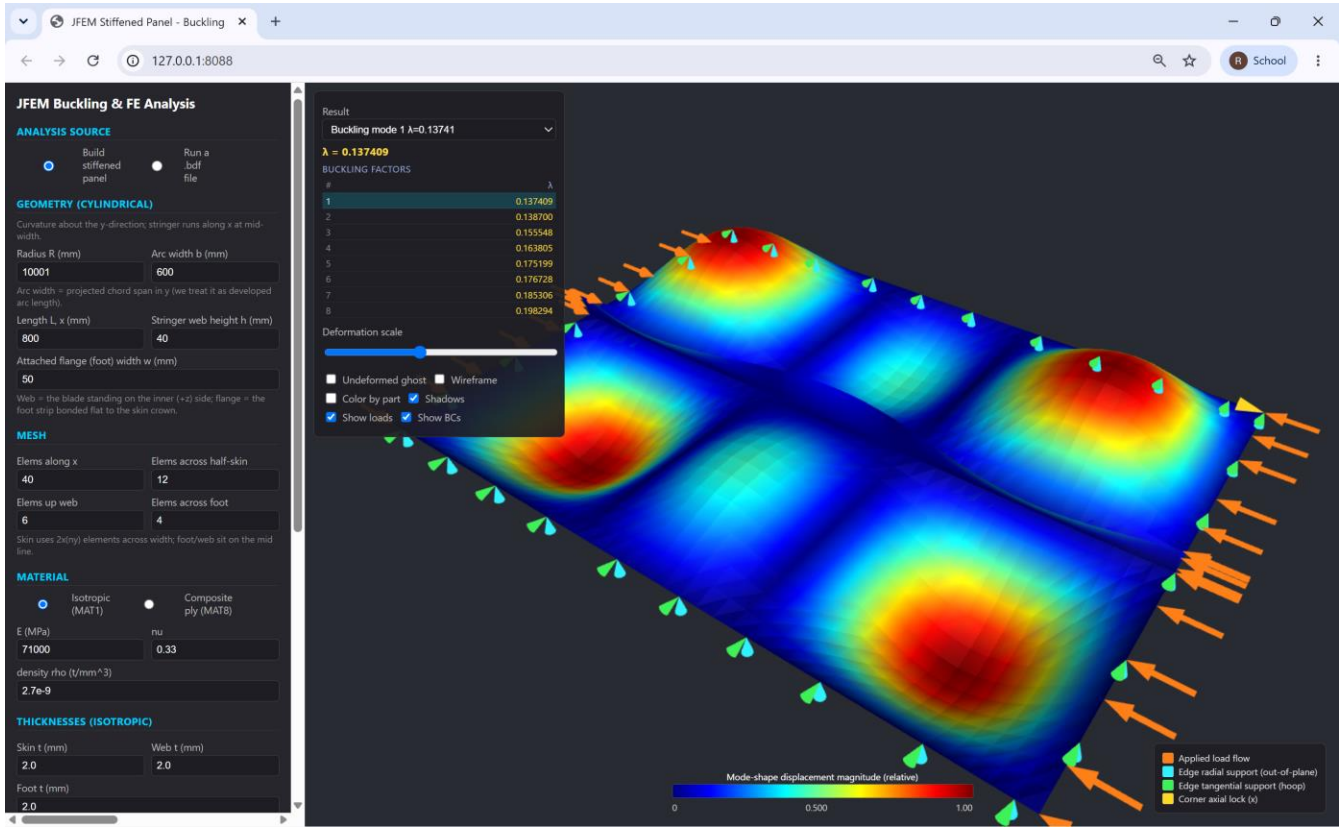


Default panel

BUCKLING RESULTS

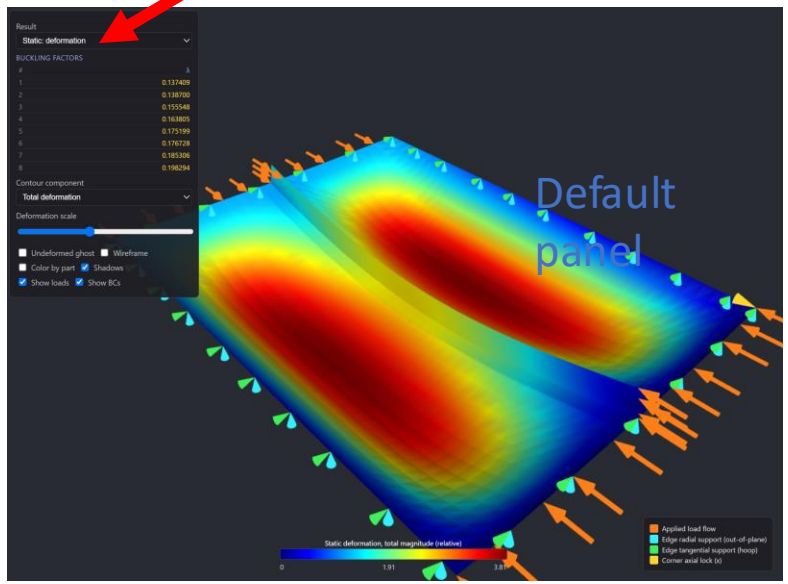
When you click the "Analyze (SOL 105)" button, the analysis will commence and a timer showing the execution time will be shown (the first execution will be slower than the following runs)

Once the analysis is completed, you can select the relevant results from the results menu. You can plot the buckling modes, static deformation (pre buckling) and stress fields.

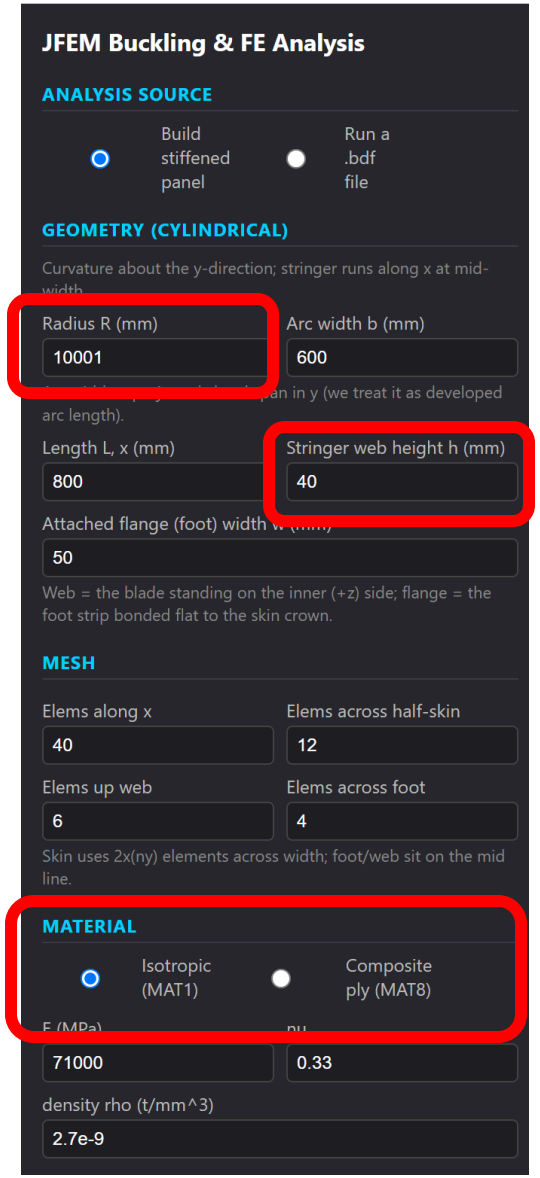


First buckling mode shown by default

Select static deformation

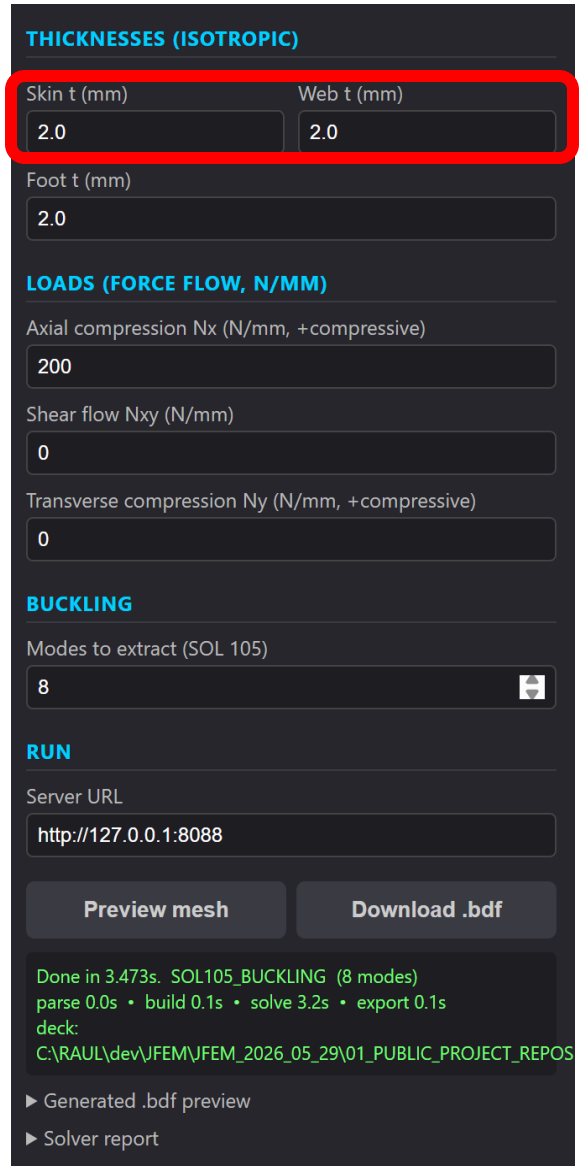


PARAMETERS AND PROPERTIES



In this exercise you will only have to change the Radius (of the cylinder to which the panel belongs, thing of a fuselage or the body of a large rocket) and the stringer height.

You will also have to select Composite for the second part of the exercise

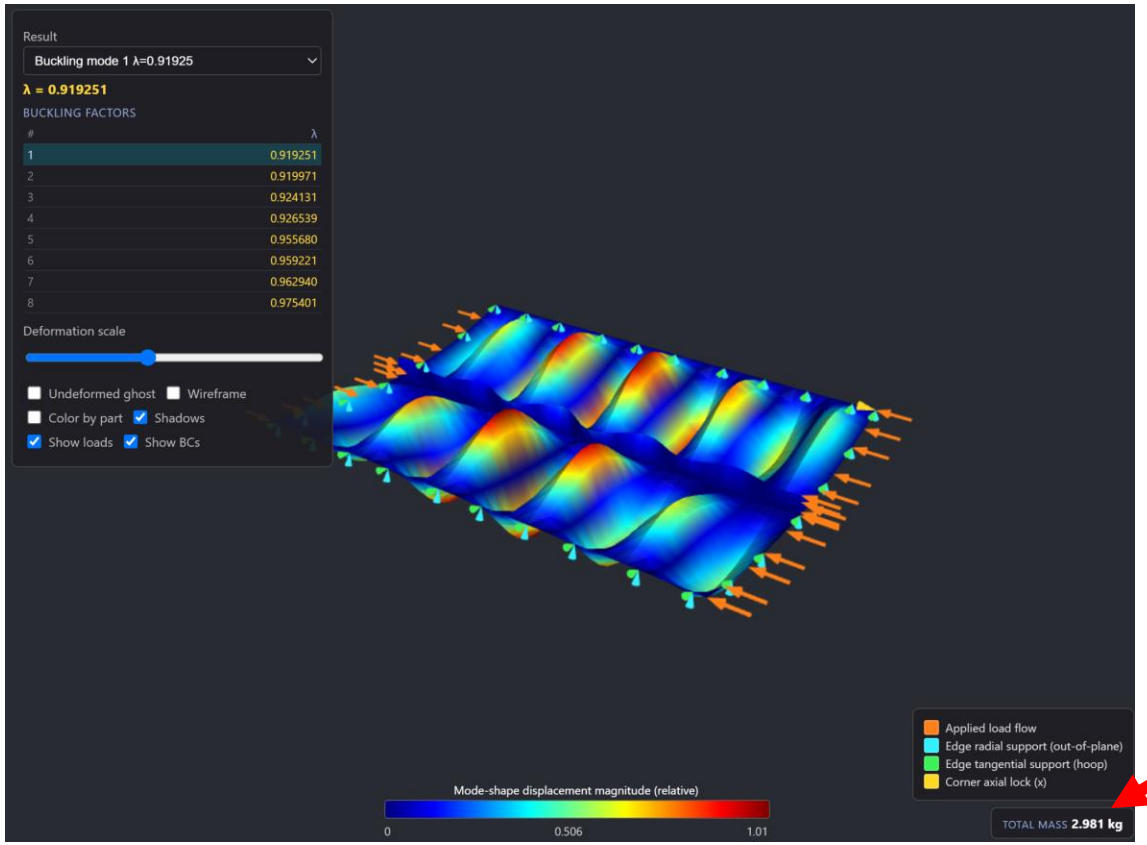


Only the skin and stringer web thicknesses (in the case of isotropic material) or the laminates will be changed

EXERCISE 1: SIZE AN ISOTROPIC CURVED PANEL

When you run the panel that appears by default you will get a result similar to the one in the image (the values will depend on the version of the OpenJFEM solver, which is still evolving).

1- Increase the thickness of the skin until you get a first eigenvalue just above 1.00. Record the thickness, the panel weight and take a screenshot of the buckling mode.



Panel weight

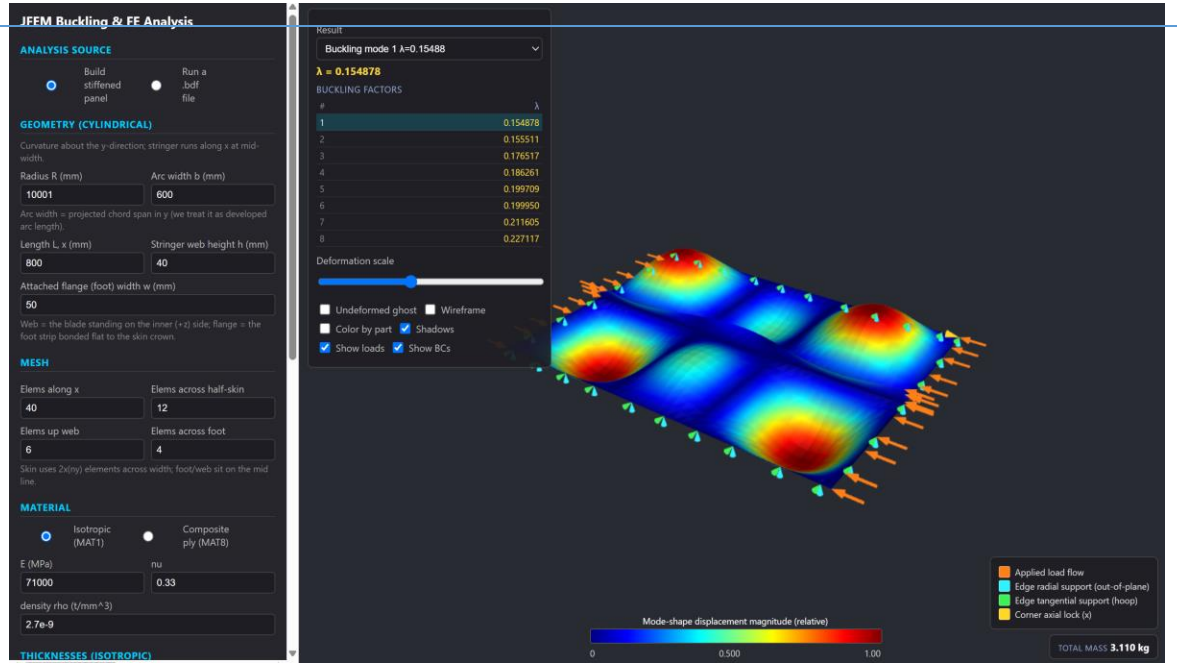
EXERCISE 2; SIZE A "FLAT" PANEL AND COMPARE WITH THE THEORY

2- Change the radius (R) of the panel to 10001mm (this makes the panel much flatter). Change the skin thickness until you get a first eigenvalue greater than 1.00 (but as close as possible to 1.00, for example 1.002)

3- Record the eigenvalue and weight for every thickness that you try and make a plot eigenvalue versus thickness. Does the trend correspond with the theory of buckling?

4- Why do you think that the flat panel is much more prone to buckling than a curved panel?

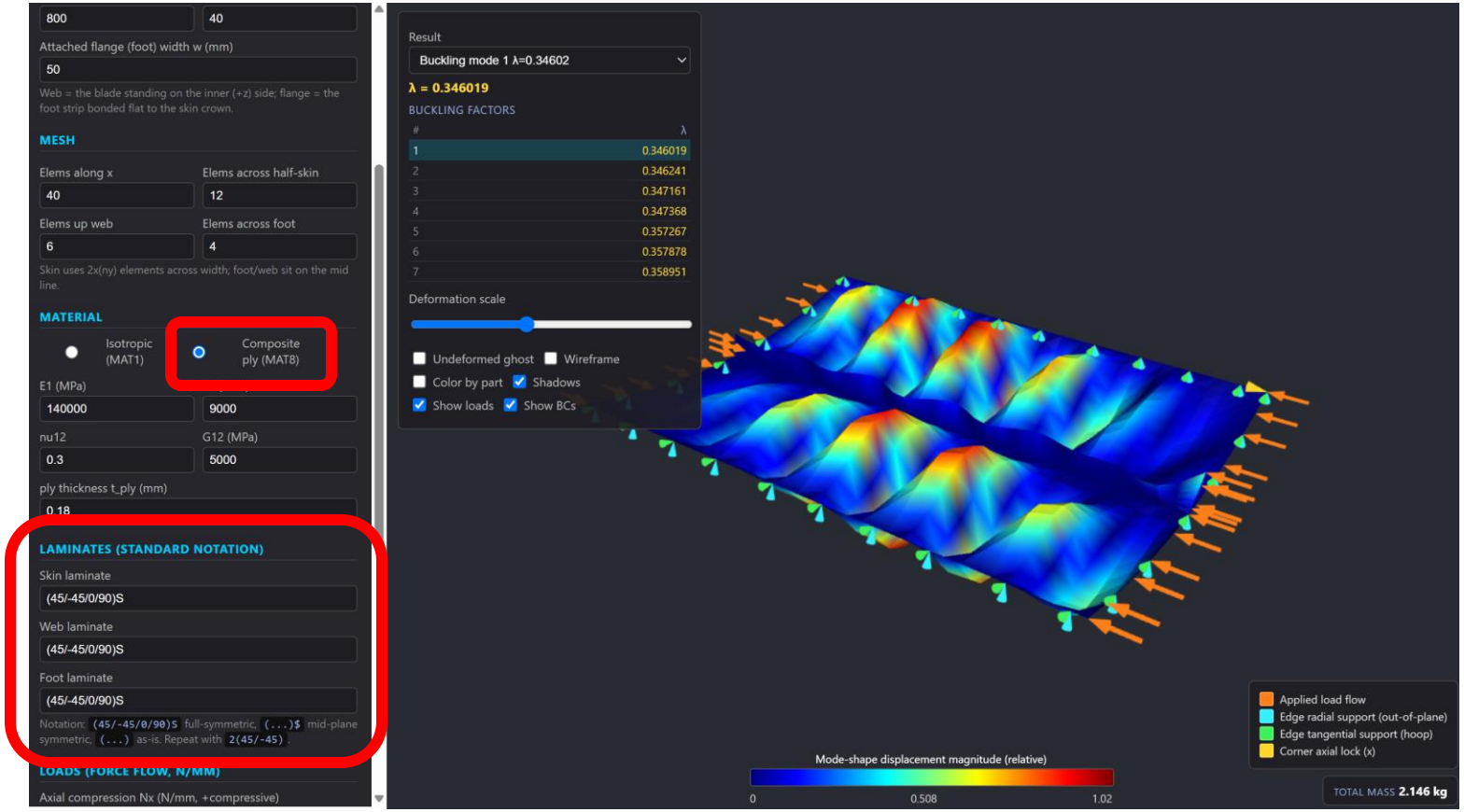
5- Reduce the stringer web height until the first global panel mode has an eigenvalue of 1.00. Is this design robust? Why?



EXERCISE 3; SIZE A FLAT COMPOSITE PANEL

Starting from the previous solution, change the material to Composite.

6- Add plies of different orientations using the notation shown until you get a first eigenvalue of just above 1.00. Record the weight. Try other laminates and make a table of the results. Which is the optimum laminate? If you get local stringer buckling, add plies to the web. If you get global buckling, increase the web height.



FINAL QUESTIONS

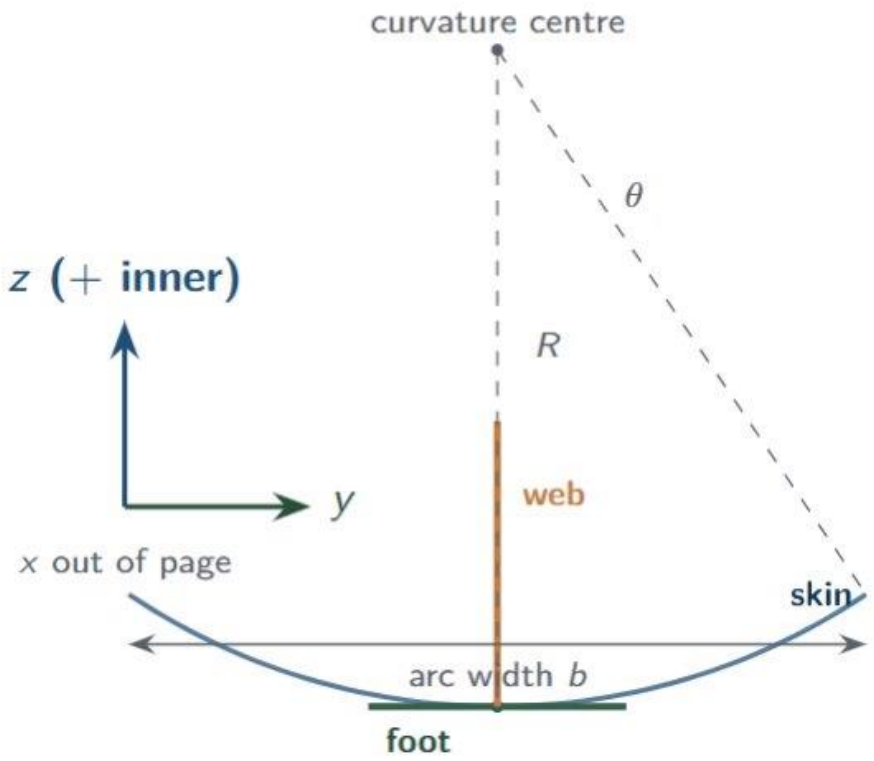
7- Are the boundary conditions of this Finite Elements Model correct? How would you change the model to make it more correct?

8- Any final comments on this exercise and recommendations for better clarity or general improvement

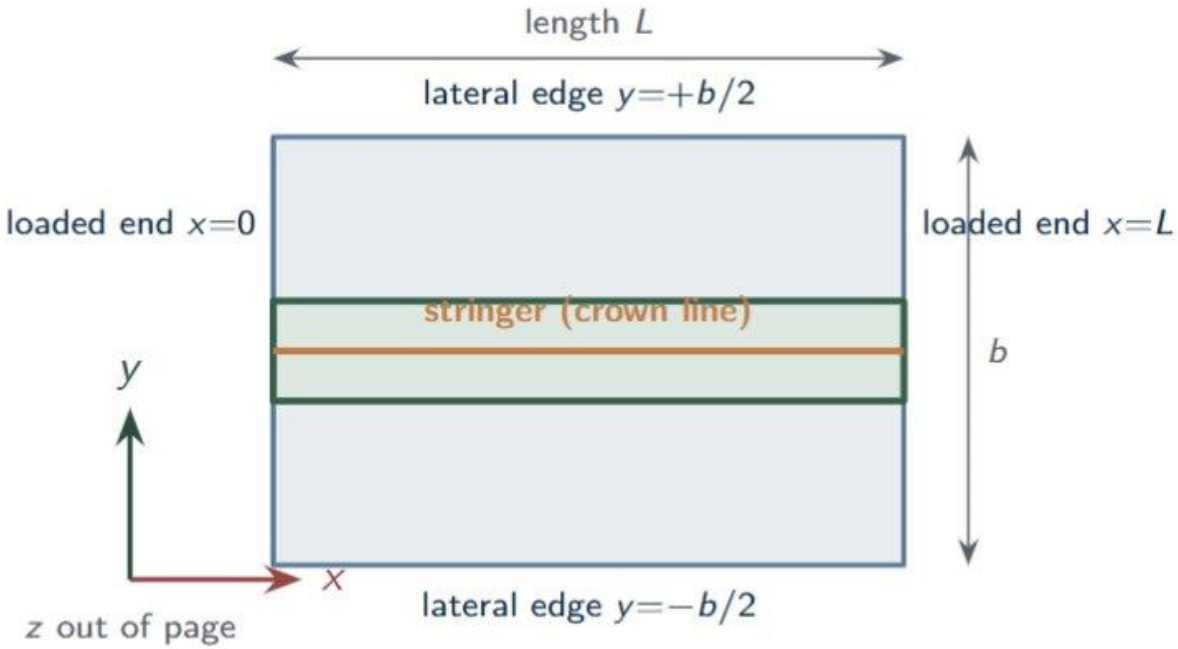
Upload a pdf file with a concise document covering all the points above.

Deadline; the day of the Final Ordinary Exam

PANEL GEOMETRY DEFINITIONS



(a) cross-section (y–z)



(b) plan (x–y)

FEM LINEAR BUCKLING THEORY

The two-step SOL 105 procedure

1. **Static preload.** Solve $\mathbf{K}\,\mathbf{u}_0 = \mathbf{f}$ for the reference load $\mathbf{f}$ (the applied force flows). This gives the membrane stress state in every element.
2. **Differential stiffness.** From that stress state, assemble the **geometric (differential) stiffness** $\mathbf{K}_g(\mathbf{u}_0)$, which captures how membrane forces stiffen or soften out-of-plane bending.
3. **Eigenproblem.** Find the load multipliers λ and mode shapes ϕ that make the tangent stiffness singular.

$$\left(\mathbf{K} + \lambda\,\mathbf{K}_g\right)\phi \;=\; \mathbf{0}, \qquad \mathbf{K}\,\phi \;=\; -\,\lambda\,\mathbf{K}_g\,\phi.$$

The smallest positive λ_1 is the **critical load factor**; ϕ_1 is the buckling shape. In the deck this is `SOL 105` with a static `SUBCASE`, `STATSUB` linking the preload, and `EIGRL` selecting the lowest modes.

- Total potential energy = strain energy − work of loads. Equilibrium is a stationary point; **stability** is whether that point is a minimum.
- The tangent stiffness is $\mathbf{K}_T(\lambda) = \mathbf{K} + \lambda\,\mathbf{K}_g$. As the load grows ($\lambda \uparrow$), membrane compression makes $\mathbf{K}_g$ subtract from $\mathbf{K}$.
- Buckling is the load at which $\mathbf{K}_T$ first loses positive-definiteness, i.e. $\det(\mathbf{K} + \lambda\,\mathbf{K}_g) = 0$.
- $\mathbf{K}_g$ is *linear in the preload stress*, so the critical loads scale as eigenvalues λ of the applied pattern.

Reading the number

- λ_1 = factor on *all* applied flows simultaneously.
- Buckling **load** = $\lambda_1 \times$ (applied N_x, N_{xy}, N_y).
- Higher modes $\lambda_2, \lambda_3, \ldots$ are further bifurcations; the first one governs.

THEORETICAL RESULTS FOR AN ISOTROPIC FLAT STIFFENED PLATE

For a simply-supported *flat* rectangular plate of thickness t under uniaxial compression, the classical critical stress is

$$\sigma_{cr} = k \frac{\pi^2 E}{12\,(1-\nu^2)} \left(\frac{t}{b}\right)^2,$$

so the critical *force flow* $N_{cr} = \sigma_{cr}\, t \propto t^3$.

This t^3 scaling is the lever you will use in the exercise: a small thickness change moves λ a lot.

Curvature & stiffeners help

- Curvature (R) raises the critical load above the flat-plate value.
- The stringer subdivides the skin and adds bending stiffness, raising λ further and changing the mode shape.
- The web can itself buckle (crippling) – watch which part moves in ϕ_1.